portion of the hyperbola as $x = \frac{4}{3}\sqrt{9+y^2}$. So, the volume of the solid generated is given by

$$V = \pi\int_1^6 x^2\,dy = \pi\int_1^6 \left(\frac{4}{3}\sqrt{9+y^2}\right)^2 dy = \pi\left(\frac{16}{9}\right)\int_1^6 \left(9+y^2\right)dy$$

$$= \pi\left(\frac{16}{9}\right)\left(9y+\frac{y^3}{3}\right)_1^6 = \pi\left(\frac{16}{9}\right)\left[(54+72)-(9+\frac{1}{3})\right] = \frac{5600}{27}\pi$$

Example 9.69

Find, by integration, the volume of the solid generated by revolving about y-axis the region bounded by the curves $y = \log x$, $y = 0$, $x = 0$ and $y = 2$.

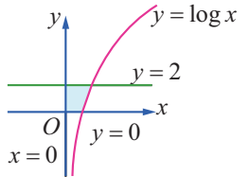


Fig. 9.45

Solution

The region to be revolved is sketched in Fig.9.45.

Since revolution is made about the y-axis, the volume of the solid generated is given by

$$V = \pi\int_0^2 x^2\,dy = \pi\int_0^2 \left(e^y\right)^2 dy = \pi\int_0^2 e^{2y}\,dy$$

$$= \pi\left[\frac{e^{2y}}{2}\right]_0^2 = \frac{\pi}{2}\left(e^4-1\right).$$

EXERCISE 9.9

1. Find, by integration, the volume of the solid generated by revolving about the x-axis, the region enclosed by $y = 2x^2$, $y = 0$ and $x = 1$.
2. Find, by integration, the volume of the solid generated by revolving about the x-axis, the region enclosed by $y = e^{-2x}$ $y = 0$, $x = 0$ and $x = 1$
3. Find, by integration, the volume of the solid generated by revolving about the y-axis, the region enclosed by $x^2 = 1 + y$ and $y = 3$.
4. The region enclosed between the graphs of $y = x$ and $y = x^2$ is denoted by R, Find the volume generated when R is rotated through $360°$ about x-axis.
5. Find, by integration, the volume of the container which is in the shape of a right circular conical frustum as shown in the Fig 9.46.
6. A watermelon has an ellipsoid shape which can be obtained by revolving an ellipse with major-axis 20 cm and minor-axis 10 cm about its major-axis. Find its volume using integration.

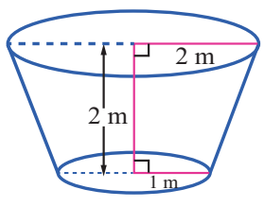


Fig. 9.46

EXERCISE 9.10

Choose the correct or the most suitable answer from the given four alternatives :

1. The value of $\int_0^{\frac{2}{3}} \frac{dx}{\sqrt{4-9x^2}}$ is

(1) $\frac{\pi}{6}$ (2) $\frac{\pi}{2}$ (3) $\frac{\pi}{4}$ (4) π

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